EEC 431

ELECTROMAGNETIC FIELD THEORY

HIGHER NATIONAL DIPLOMA (HND) II

LECTURE NOTES

Organised by:

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COURSE BRIEF

Electromagnetic field theory is concerned with the study of charges at rest and in motion. Electromagnetic principles are fundamental to the study of electrical engineering. Electromagnetic theory is also required for the understanding, analysis and design of various electrical, electromechanical and electronic systems.

Electromagnetic theory can be thought of as generalization of circuit theory. Electromagnetic theory deals directly with the electric and magnetic field vectors whereas circuit theory deals with the voltages and currents. Voltages and currents are integrated effects of electric and magnetic fields respectively.

Electromagnetic field problems involve three space variables along with the time variable and hence the solution tends to become correspondingly complex. Vector analysis is the required mathematical tool with which electromagnetic concepts can be conveniently expressed and best comprehended. Since use of vector analysis in the study of electromagnetic field theory is prerequisite, first we will go through vector algebra.

Applications of Electromagnetic theory:

This module basically consists of static electric fields, static magnetic fields, time-varying fields & it's applications.

One of the most common applications of electrostatic fields is the deflection of a charged particle such as an electron or proton in order to control its trajectory. The deflection is achieved by maintaining a potential difference between a pair of parallel plates. This principle is used in CROs, ink-jet printer etc. Electrostatic fields are also used for sorting of minerals for example in ore separation. Other applications are in electrostatic generator and electrostatic voltmeter. The most common applications of static magnetic fields are in dc machines. Other applications include magnetic deflection, magnetic separator, cyclotron, hall effect sensors, magneto hydrodynamic generator etc.

Recommended Textbooks:

[1]. W. H. Hayt (Jr), J. A. Buck, “Engineering Electromagnetics”, TMH

[2]. K. E. Lonngren, S.V. Savor, “Fundamentals of Electromagnetics with Matlab”, PHI

[3]. E.C.Jordan, K.G. Balmain, “Electromagnetic Waves & Radiating System”, PHI.

[4]. M. N. Sadiku, “Elements of Electromagnetics”, Oxford University Press.

[5] D. Tougaw “Applied Electromagnetic Field Theory” Valparaiso University Press

INTRODUCTION

Vector Analysis:

The quantities that we deal in electromagnetic theory may be either **scalar** or **vectors.** Scalars are quantities characterized by magnitude only. A quantity that has direction as well as magnitude is called a vector. In electromagnetic theory both scalar and vector quantities are function of *time* and *position*.

A vector $\vec{A}$ can be written as, $\vec{A} = \hat{a}\, A$, where, $A = |\vec{A}|$ is the magnitude and $\hat{a} = \frac{\vec{A}}{|A|}$ is the unit vector which has unit magnitude and same direction as that of $\vec{A}$.

Two vectors $\vec{A}$ and $\vec{B}$ are added together to give another vector $\vec{C}$. Thus,

$$\vec{C} = \vec{A} + \vec{B} \qquad (1.1)$$

Let us see the manipulations (for addition) of two vectors, which has two rules: the **Parallelogram law** and **the Head & tail rule.**

Scaling of a vector is defined as $\vec{C} = \alpha\vec{B}$, where $\vec{C}$ is scaled version of vector $\vec{B}$ and α is a scalar. Some important laws of vector algebra are:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \quad \ldots \quad \textit{(Commutative Law)} \qquad (1.3)$$

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} \quad \ldots. \quad \textit{(Associative Law)} \qquad (1.4)$$

$$\alpha(\vec{A} + \vec{B}) = \alpha\vec{A} + \alpha\vec{B} \quad \ldots. \textit{(Distributive Law)} \qquad (1.5)$$

The position vector $\vec{r_Q}$ of a point P is the directed distance from the origin (O) to P,

i.e., $\vec{r_Q}$ = $\overrightarrow{OP}$. If $\vec{r_Q}$ = OP and $\vec{r_P}$ = OQ are the position vectors of the points P and Q then the distance vector

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \vec{r_P} - \vec{r_Q}$$

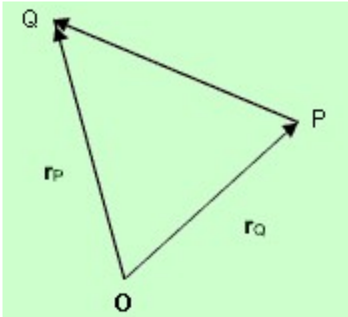


Fig 1.3: Distance Vector

Product of Vectors

When two vectors $\vec{A}$ and $\vec{B}$ are multiplied, the result is either a scalar or a vector depending how the two vectors were multiplied. The two types of vector multiplication are:

Scalar product (or dot product) $\vec{A}\cdot\vec{B}$ gives a scalar.

Vector product (or cross product) $\vec{A}\times\vec{B}$ gives a vector.

The dot product between two vectors is defined as $\vec{A}\cdot\vec{B} = |A||B|cos\theta_{AB}$ ....(1.6)

Vector product $$\vec{A}\times\vec{B} = |A||B|\sin\theta_{AB}\cdot\vec{n}$$

$\vec{n}$ is unit vector perpendicular to $\vec{A}$ and $\vec{B}$

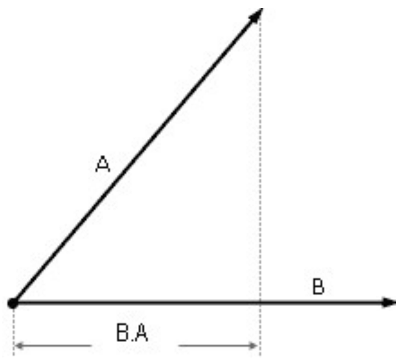


Fig 1.4 : Vector dot product

The dot product is commutative i.e., $\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A}$ and distributive i.e., $\vec{A}\cdot\left(\vec{B}+\vec{C}\right) = \vec{A}\cdot\vec{B} + \vec{A}\cdot\vec{C}$

Associative law does not apply to scalar product.

The vector or cross product of two vectors $\vec{A}$ and $\vec{B}$ is denoted by $\vec{A}\times\vec{B}$. $\vec{A}\times\vec{B}$ is denoted by perpendicular to the plane containing $\vec{A}$ and $\vec{B}$, the magnitude is given by $|A||B|\sin\theta_{AB}$ and direction is given by right hand rule.

$$\vec{A}\times\vec{B} = \hat{a}_n AB\sin\theta_{AB} \tag{1.7}$$

where $\hat{a}_n$ is the unit vector given by,

$$\hat{a}_n = \frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}.$$

The following relations hold for vector product.

$$\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$$ i.e., cross product is non-commutative (1.8)

$$\vec{A}\times(\vec{B}+\vec{C}) = \vec{A}\times\vec{B}+\vec{A}\times\vec{C}$$ i.e., cross product is distributive (1.9)

$$\vec{A}\times(\vec{B}\times\vec{C}) \neq (\vec{A}\times\vec{B})\times\vec{C}$$ i.e., cross product is non associative (1.10)

<u>Scalar and vector triple product:</u>

Scalar triple product $\vec{A}\cdot(\vec{B}\times\vec{C}) = \vec{B}\cdot(\vec{C}\times\vec{A}) = \vec{C}\cdot(\vec{A}\times\vec{B})$ (1.11)

Vector triple product $\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{A}\cdot\vec{C}) - \vec{C}(\vec{A}\cdot\vec{B})$ (1.12)

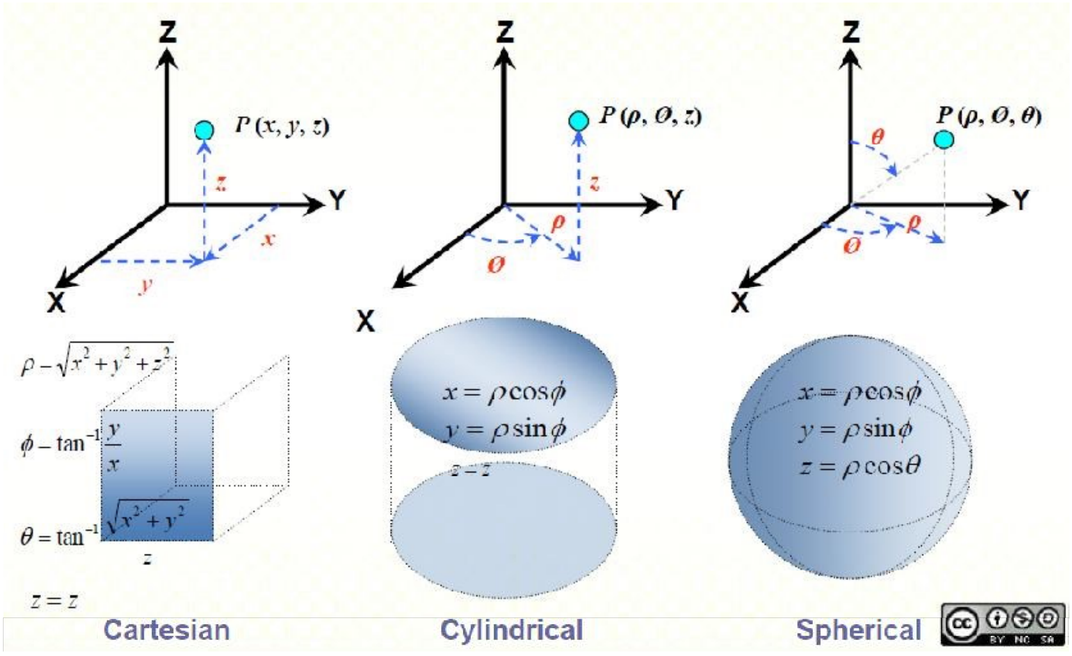
<u>Co-ordinate Systems:</u>

In order to describe the spatial variations of the quantities, we require using appropriate co-ordinate system. A point or vector can be represented in an orthogonal coordinate system. An orthogonal system is one in which the co-ordinates are mutually perpendicular.

In electromagnetic theory many physical quantities are vectors, which are having different components. So we use orthogonal co-ordinate systems for representing those quantities and depending on the symmetry of the physical quantities different coordinate systems are used.

The electromagnetic (static) problem cases

Coordinate	The problems most often encountered	
	Electrostatic Cases	**Magnetostatic Cases**
Cartesian	- Any uniform charge distribution using a scale of cartesian coordinates.	- The current flowing on the horizontal infinite extent plane.
Cylindrical	- Uniform charge distribution on cylindrical conductor. - Uniform charge distribution on the infinite length of line. - Uniform charge distribution on the infinite extent horizontal plane. - Uniform charge distribution on the horizontal circle plane. - Uniform charge distribution on the circle line.	- The current flowing in circumference. - The current flowing in an infinite straight line. - The current flows in the solenoid and toroid.
Spherical	- Uniform charge distribution on the surface of a sphere. - Uniform charge distribution of the point.	- Problem for the case of spheres less found.



Del Operator:

Del is a vector differential operator. The del operator will be used in for differential operations throughout any course on field theory. The following equation is the del operator for different coordinate systems.

$$\nabla = \frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z = \nabla_{x,y,z}$$

$$\nabla = \frac{\partial}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial}{\partial \phi}\hat{a}_\phi + \frac{\partial}{\partial z}\hat{a}_z$$

$$\nabla = \frac{\partial}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\hat{a}_\phi$$

Gradient of a Scalar:

- The gradient of a scalar field, V, is a vector that represents both the magnitude and the direction of the maximum space rate of increase of V.

$$\nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z = \nabla V_{x,y,z}$$

- To help visualize this concept, take for example a topographical map. Lines on the map represent equal magnitudes of the scalar field. The gradient vector crosses map at the location where the lines packed into the most dense space and perpendicular (or normal) to them. The orientation (up or down) of the gradient vector is such that the field is increased in magnitude along that direction.

Fundamental properties of the gradient of a scalar field

– The magnitude of gradient equals the maximum rate of change in V per unit distance

– Gradient points in the direction of the maximum rate of change in V

– Gradient at any point is perpendicular to the constant V surface that passes through that point – The projection of the gradient in the direction of the unit vector **a**, is

$$\nabla V \cdot \hat{a}$$

and is called the directional derivative of V along **a**. This is the rate of change of V in the direction of **a**.

– If **A** is the gradient of V, then V is said to be the scalar potential of **A.**

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$$\nabla = \frac{\partial}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial}{\partial \phi}\hat{a}_\phi + \frac{\partial}{\partial z}\hat{a}_z$$

$$\nabla = \frac{\partial}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\hat{a}_\phi$$

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Divergence of a Vector:

- The divergence of a vector, **A**, at any given point P is the outward flux per unit volume as volume shrinks about P.

$$div\vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta v \to 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

Divergence Theorem:

- The divergence theorem states that the total outward flux of a vector field, **A**, through the closed surface, S, is the same as the volume integral of the divergence of **A**.
- This theorem is easily shown from the equation for the divergence of a vector field.

$$\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$$

$$div\vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta v \to 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

$$\int_v \nabla \cdot \vec{A} dv = \oint_s \vec{A} \cdot d\vec{S}$$

Curl of a Vector:

- The curl of a vector, **A** is an axial vector whose magnitude is the maximum circulation of **A** per unit area as the area tends to zero and whose direction is the normal direction of the area when the area is oriented to make the circulation maximum.

-Curl of a vector in each of the three primary coordinate systems are,

Cartesian
$$\nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right]\hat{a}_x - \left[\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z}\right]\hat{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right]\hat{a}_z$$

Cylindrical
$$\nabla \times \vec{A} = \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix} = \left[\frac{1}{\rho}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right]\hat{a}_\rho - \left[\frac{\partial A_z}{\partial \rho} - \frac{\partial A_\rho}{\partial z}\right]\hat{a}_\phi + \frac{1}{\rho}\left[\frac{\partial(\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi}\right]\hat{a}_z$$

Spherical
$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

$$\nabla \times \vec{A} = \frac{1}{r\sin\theta}\left[\frac{\partial(A_\phi \sin\theta)}{\partial \theta} - \frac{\partial A_\theta}{\partial \phi}\right]\hat{a}_r - \frac{1}{r}\left[\frac{\partial(rA_\phi)}{\partial r} - \frac{1}{\sin\theta}\frac{\partial A_r}{\partial \phi}\right]\hat{a}_\theta + \frac{1}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta}\right]\hat{a}_\phi$$

Stokes Theorem:

Stokes theorem states that the circulation of a vector field A, around a closed path, L is equal to the surface integral of the curl of A over the open surface S bounded by L. This theorem has been proven to hold as long as **A** and the curl of **A** are continuous along the closed surface S of a closed path L

- This theorem is easily shown from the equation for the curl of a vector field.

$$\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$$

$$curl\vec{A} = \nabla \times \vec{A} = \left(\lim_{\Delta S \to 0} \frac{\oint_L \vec{A} \cdot d\vec{l}}{\Delta S} \right)_{max} \hat{a}_n$$

$$\oint_L \vec{A} \cdot d\vec{l} - \oint_s \left(\nabla \times \vec{A}\right) \cdot d\vec{S}$$

Examples: 1

$$\boldsymbol{A} = \hat{a}_x + 2\hat{a}_y + 3\hat{a}_z$$

1. Given that

$$\boldsymbol{B} = -2\hat{a}_x + \hat{a}_y$$

a. Determine the angle between the vectors ***A*** and ***B*** .

b. Find the unit vector which is perpendicular to both ***A*** and ***B*** .

Solution:

a. We know that for two given vector A and B,

$$\boldsymbol{A}.\boldsymbol{B} = |A|.|B|\cos\theta$$

For the two vectors A and B

$$|A| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

$$|B| = \sqrt{(-2)^2 + 1} = \sqrt{5}$$

$$\therefore \sqrt{14}.\sqrt{5}\cos\theta = 1 \text{ or } \theta = \cos^{-1}\left(\frac{1}{\sqrt{70}}\right)$$

b. We know that is perpendicular to both A and B.

$$\therefore \boldsymbol{A}\times\boldsymbol{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ 1 & 2 & 3 \\ -2 & 0 & 1 \end{vmatrix}$$

$$= 2\hat{a}_x - (1+6)\hat{a}_4 + 4\hat{a}_3$$

The unit vector n perpendicular to both ***A*** and ***B*** is given by,

$$\hat{n} = \frac{\boldsymbol{A}\times\boldsymbol{B}}{|\boldsymbol{A}\times\boldsymbol{B}|} = \frac{2\hat{a}_x - 7\hat{a}_y + 4\hat{a}_z}{\sqrt{2^2+(-7)^2+4^2}} = \frac{2\hat{a}_x - 7\hat{a}_y + 4\hat{a}_z}{\sqrt{69}}$$

Examples: 2

Given the vectors

$$\boldsymbol{A} = \hat{a}_x + 2\hat{a}_y + 5\hat{a}_z$$

$$\boldsymbol{B} = 5\hat{a}_\rho - \hat{a}_\phi + 3\hat{a}_z$$

Find:

a. The vector ***C*** = ***A*** + ***B*** at a point P (0, 2,-3).

b. The component of ***A*** along ***B*** at P.

Solution:

The vector B is cylindrical coordinates.

This vector in Cartesian coordinate can be written as:

$$\boldsymbol{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

Where

$$B_x = \boldsymbol{B}.\hat{a}_x = 5\hat{a}_\rho.\hat{a}_x - \hat{a}_\phi.\hat{a}_x + 3\hat{a}_3.\hat{a}_x$$

$$= 5\cos\phi + \sin\phi$$

$$B_y = \boldsymbol{B}.\hat{a}_y = 5\sin\phi - \cos\phi$$

$$B_z = \boldsymbol{B}.\hat{a}_z$$

The point P(0,2,-3) is in the y-z plane for which $\phi = \frac{\pi}{2}$.

$$\therefore \boldsymbol{B} = \hat{a}_x + 5\hat{a}_y + 3\hat{a}_z$$

a. ***C = A + B***

$$= \left(\hat{a}x + 2\hat{a}_y + 5\hat{a}_z\right) + \left(\hat{a}x + 5\hat{a}_y + 3\hat{a}_z\right)$$

$$= 2\hat{a}x + 7\hat{a}_y + 8\hat{a}_z$$

b. Component of ***A*** along ***B*** is $|A|\cos\theta$ θ where is the angle between A is and B.

i.e., $\frac{\boldsymbol{A}.\boldsymbol{B}}{|B|}$ $\frac{1+10+15}{\sqrt{1+25+9}} = \frac{26}{\sqrt{35}}$ =

Examples: 3

A vector field is given by

$$\boldsymbol{A} = \rho\cos\phi\hat{a}_\rho + \rho z\sin\phi\hat{a}_z$$

Transform this vector into rectangular co-ordinates and calculate its magnitude at P(1,0,1).

Solution:

Given, $\boldsymbol{A} = \rho\cos\phi\hat{a}_\rho + \rho z\sin\phi\hat{a}_z$

The components of the vector in Cartesian coordinates can be computed as follows:

$$A_x = \boldsymbol{A}.\hat{a}_x$$

$$= \rho\cos\phi\cos\phi = \rho\cos^2\phi = \sqrt{x^2+y^2}.\left(\frac{x}{\sqrt{x^2+y^2}}\right)$$

$$= \frac{x^2}{\sqrt{x^2+y^2}}$$

$$A_y = \boldsymbol{A}\,\hat{a}_y$$

$$= \rho \cos\phi \sin\phi = \sqrt{x^2 + y^2} . \frac{x}{\sqrt{x^2 + y^2}} . \frac{y}{\sqrt{x^2 + y^2}}$$

$$= \frac{xy}{\sqrt{x^2 + y^2}}$$

$$A_z = \sqrt{x^2 + y^2} . z \frac{y}{\sqrt{x^2 + y^2}} = yz$$

$$\therefore \boldsymbol{A} = \frac{x^2}{\sqrt{x^2 + y^2}} \hat{a}_x + \frac{xy}{\sqrt{x^2 + y^2}} \hat{a}_y + yz\, \hat{a}_z$$

$$\boldsymbol{A}\Big|_{(1,0,1)} = \frac{1}{\sqrt{1}} \hat{a}_x + 0 + 0$$

$$= \hat{a}_x$$

$$\therefore \left|\boldsymbol{A}\right|\Big|_{(1,0,1)} = 1$$

Examples: 4

A given vector function is defined by $\boldsymbol{F} = y\hat{a}_x + x\hat{a}_y$. Evaluate the scalar line integral from a point

P1(1, 1, -1) to P2(2, 4, -1).

a. along the parabola $y = x^2$

b. along the line joining the two points.

Is ***F*** a conservative field?

Solution:

$$\boldsymbol{F} = y\hat{a}_x + x\hat{a}_y$$

$$d\boldsymbol{l} = dx\,\hat{a}_x + dy\,\hat{a}_y$$

$$\therefore \boldsymbol{F}.d\boldsymbol{l} = ydx + xdy$$

a. For evaluating the line integral along the parabola $y = x^2$, we find that

$$dy = 2x\,dx$$

$$\therefore \int_{P_1}^{P_2} F \cdot dl = \int_1^2 x^2 dx + 2x^2 dx$$

$$= \int_1^2 3x^2 dx = \left[x^3\right]_1^2 = z$$

b. In this case we observe that z1 = z2 = -1, hence the line joining the points P1 and P2 lies in the z = -1 plane and can be represented by the equation

$$y - 1 = \frac{4-1}{2-1}(x-1)$$

Or, $y = 3x - 2$ $\therefore dy = 3\,dx$

$$\boldsymbol{F} \cdot dl = (3x - 2)dx + x \cdot 3dx$$

$$= (6x - 2)dx$$

$$\therefore \int_{P_1}^{P_2} \boldsymbol{F} \cdot d\boldsymbol{l} = \int_1^2 (6x-2)\,dx$$

$$= \left[\frac{6x^2}{2} - 2x\right]_1^2$$

$$= [12 - 4 - 3 + 2]$$

$$= 7$$

The field ***F*** is a conservative field.

Examples: 5.

If $\boldsymbol{D} = \frac{1}{r}\hat{a}_r$, calculate $\int_S \boldsymbol{D} \cdot d\boldsymbol{S}$ over a hemispherical surface bounded by r =2 & $0 \le \theta \le \pi/2$.

Solution:

In spherical polar coordinates

$$d\boldsymbol{S} = r^2 \sin\theta\, d\theta\, d\phi\, \hat{a}_r$$

$$\therefore \int_S \boldsymbol{D} \cdot d\boldsymbol{S} = \int_0^2 \int_0^{\pi/2} 2 . \sin\theta\, d\theta\, d\phi$$

$$= 4\pi \int_0^{2\pi} \sin\theta\, d\theta$$

$$= 4\pi$$

HND II: Electrical & Electronics Engineering
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TOPIC 1: Principles & applications of Electrostatics

PRINCIPLES & APPLICATIONS OF ELECTROSTATICS

Coulombs' LAW

The law expresses the force between two charges Q_1,and Q_2 as shown in Fig. 1.0 below. If these charges are positive, the force is repulsive.

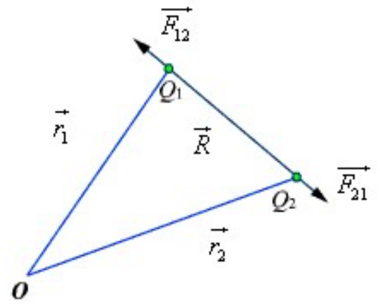


Figure 1.0: Two-points charges Q_1,and Q_2

The law states that the force between two-point charges Q_1,and Q_2separated in vacuum or free space by a distance R which is large compared to their size… is directly proportional to the product of their charges and inversely proportional to the square of the distance between them. This force acts along the line joining the two charges. Mathematically:

$$F = \frac{kQ_1Q_2}{R^2} \quad (1)$$

Where:

Q_1 and Q_2: point charge expressed in Coulombs(C)

F force expressed in Newtons (*N*)

k is the constant proportionality expressed as $k = \frac{1}{4\pi\varepsilon_o}$

ε_o is constant called the permittivity of free space it's magnitude = 8.85$\times 10^{-12}$= $\frac{1}{36\pi} \times 10^{-9}$F/m

Force on a Point Charge Place on External Field

Note that, we assumed the charges are placed in a free space. However, if the charges are any other dielectric medium, then the relative permittivity of that dielectric medium or the dielectric constant of the medium must be used.

Thus, the above equation can be rewritten as

$\varepsilon = \varepsilon_0 \varepsilon_r$ instead where ε_r

$$F = \frac{1}{4\pi\varepsilon_0}\frac{Q_1Q_2}{R^2} \quad (2)$$

As shown in the Fig. 1.0 above, let the position vectors of the point charges $Q_1$and Q_2 are given by $\vec{r_1}$ and $\vec{r_2}$. Let $\vec{F_{12}}$ represent the force on Q_1 due to charge Q_2. Thus, the charges are separated by a distance of $R = \left|\vec{r_1} - \vec{r_2}\right| = \left|\vec{r_2} - \vec{r_1}\right|$. Then we can define the unit vectors as

$$\widehat{a_{12}} = \frac{\left(\vec{r_2} - \vec{r_1}\right)}{R} \quad \text{and} \quad \widehat{a_{21}} = \frac{\left(\vec{r_1} - \vec{r_2}\right)}{R}$$

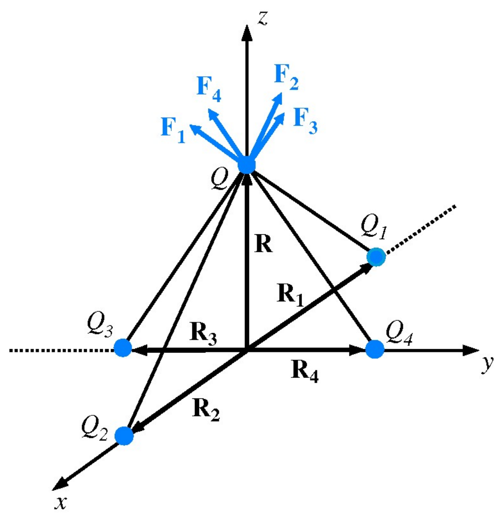
$\vec{F_{12}}$ can be defined as $\vec{F_{12}} = \frac{Q_1Q_2}{4\pi\varepsilon_0 R^2}\widehat{a_{12}} = \frac{Q_1Q_2}{4\pi\varepsilon_0 R^2}\frac{\left(\vec{r_2} - \vec{r_1}\right)}{\left|\vec{r_2} - \vec{r_1}\right|^3}$.

Numerical Example: Two identical charges are located on the *x*-axis at $x = 3$ and $x = 7$. At what point in space is the net electric field zero?

Solution: Since both charges are on the *x*-axis, the point at which the fields due to the two charges can cancel has to lie on the x-axis also. Intuitively, since the two charges are identical, that point is midway between them at $(5,0,0)$.

Example: Four charges of 10 μC each are located in free space at points with Cartesian coordinates $(-3,0,0)$, $(3,0,0)$, $(0, -3,0)$, and $(0,3,0)$. Find the force on a 20-μC charge located at $(0,0,4)$. All distances are in meters.

Solution:



$R_1 = -x\ 3$
$R_2 = x\hat{}\ 3$
$R_3 = -y\hat{}\ 3$
$R_4 = y\hat{}\ 3$
$R = z\hat{}4$

$$\mathbf{F}_1 = \frac{QQ_1}{4\pi\varepsilon_0}\frac{\mathbf{R}-\mathbf{R}_1}{|\mathbf{R}-\mathbf{R}_1|^3} = \frac{QQ_1}{4\pi\varepsilon_0}\frac{\hat{\mathbf{z}}4+\hat{\mathbf{x}}3}{125} = \frac{QQ_1}{500\pi\varepsilon_0}(\hat{\mathbf{z}}4+\hat{\mathbf{x}}3)$$
$$\mathbf{F}_2 = \frac{QQ_2}{4\pi\varepsilon_0}\frac{\mathbf{R}-\mathbf{R}_2}{|\mathbf{R}-\mathbf{R}_2|^3} = \frac{QQ_2}{4\pi\varepsilon_0}\frac{\hat{\mathbf{z}}4-\hat{\mathbf{x}}3}{125} = \frac{QQ_2}{500\pi\varepsilon_0}(\hat{\mathbf{z}}4-\hat{\mathbf{x}}3)$$
$$\mathbf{F}_3 = \frac{QQ_3}{4\pi\varepsilon_0}\frac{\mathbf{R}-\mathbf{R}_3}{|\mathbf{R}-\mathbf{R}_3|^3} = \frac{QQ_3}{4\pi\varepsilon_0}\frac{\hat{\mathbf{z}}4+\hat{\mathbf{y}}3}{125} = \frac{QQ_3}{500\pi\varepsilon_0}(\hat{\mathbf{z}}4+\hat{\mathbf{y}}3)$$
$$\mathbf{F}_4 = \frac{QQ_4}{4\pi\varepsilon_0}\frac{\mathbf{R}-\mathbf{R}_4}{|\mathbf{R}-\mathbf{R}_4|^3} = \frac{QQ_4}{4\pi\varepsilon_0}\frac{\hat{\mathbf{z}}4-\hat{\mathbf{y}}3}{125} = \frac{QQ_4}{500\pi\varepsilon_0}(\hat{\mathbf{z}}4-\hat{\mathbf{y}}3)$$

$$\mathbf{F} = \mathbf{F}_1+\mathbf{F}_2+\mathbf{F}_3+\mathbf{F}_4$$
$$= \frac{200\times10^{-12}}{500\pi\varepsilon_0}(\hat{\mathbf{z}}16) = \hat{\mathbf{z}}\frac{32\times10^{-12}}{5\pi\times8.85\times10^{-12}} = \hat{\mathbf{z}}0.23 \quad \text{(N).}$$

<u>Intensity of Electric Field</u>

The electric field intensity E (or the electric field strength) at a point is defined as the force per unit charge. This can be expressed mathematically as

$$\vec{E} = \lim_{Q\to0}\frac{\vec{F}}{Q} \quad \text{or,} \quad \vec{E} = \frac{\vec{F}}{Q} \tag{3}$$

The electric field intensity *E* at a point *r* (observation point) due a point charge *Q* located at $\vec{r}\,'$ (source point) is given by:

$$\vec{E} = \frac{Q(\vec{r}-\vec{r}\,')}{4\pi\varepsilon_0\left|\vec{r}-\vec{r}\,'\right|^3} \tag{4}$$

For a collection of *N* point charges Q_1, Q_2,.........Q_N located at $\vec{r}_1$, $\vec{r}_2$,...... $\vec{r}_N$, the electric field intensity at point $\vec{r}\,'$ is obtained as

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N}\frac{Q_k(\vec{r}-\vec{r}_i)}{\left|\vec{r}-\vec{r}_i\right|^3} \tag{5}$$

The expression (5) can be modified suitably to compute the electric filed due to a continuous distribution of charges.

For an elementary charge $dQ = \rho(\vec{r'})dv'$ i.e., considering this charge as point charge, we can write the field expression as:

$$d\vec{E} = \frac{dQ(\vec{r}-\vec{r'})}{4\pi\varepsilon_0\left|\vec{r}-\vec{r'}\right|^3} = \frac{\rho(\vec{r'})dv'(\vec{r}-\vec{r'})}{4\pi\varepsilon_0\left|\vec{r}-\vec{r'}\right|^3} \quad (6)$$

When this expression is integrated over the source region, we get the electric field at the point *P* due to this distribution of charges. Thus, the expression for the electric field at *P* can be written as:

$$\overrightarrow{E(r)} = \int_V \frac{\rho(\vec{r'})(\vec{r}-\vec{r'})}{4\pi\varepsilon_0\left|\vec{r}-\vec{r'}\right|^3}dv' \quad (7)$$

Similar technique can be adopted when the charge distribution is in the form of a line charge density or a surface charge density.

$$\overrightarrow{E(r)} = \int_L \frac{\rho_L(\vec{r'})(\vec{r}-\vec{r'})}{4\pi\varepsilon_0\left|\vec{r}-\vec{r'}\right|^3}dl' \quad (8)$$

$$\overrightarrow{E(r)} = \int_S \frac{\rho_s(\vec{r'})(\vec{r}-\vec{r'})}{4\pi\varepsilon_0\left|\vec{r}-\vec{r'}\right|^3}ds' \quad (9)$$

Numerical Example: A square plate in the *x*–*y* plane is situated in the space defined by −3 m ≤ *x* ≤ 3 m and −3 m ≤ *y* ≤ 3 m. Find the total charge on the plate if the surface charge density is given by $\rho_s = 4y^2$ (μC/m^2).

Solution:

$$\begin{aligned}
\rho_s &= 4y^2 \\
Q &= \int_S \rho_s \, ds \\
&= \int_{-3}^{3}\int_{-3}^{3} 4y^2 \, dx \, dy \\
&= \left.\left.\frac{4y^3x}{3}\right|_{-3}^{3}\right|_{-3}^{3} = 432\ \mu\text{C} = 0.432 \quad (\text{mC}).
\end{aligned}$$

Electric flux density:

As stated earlier electric field intensity or simply 'Electric field' gives the strength of the field at a particular point. The electric field depends on the material media in which the field is being considered. The flux density vector is defined to be independent of the material media (as we'll see that it relates to the charge that is producing it). For a linear isotropic medium under consideration, the flux density vector is defined as:

$$\vec{D} = \varepsilon \vec{E} \qquad (10)$$

We define the electric flux ψ as

$$\psi = \int_S \vec{D} . d\vec{s} \qquad (11)$$

Gauss's LAW

Gauss's law is one of the fundamental laws of electromagnetism and **it states that**…

the total electric flux through a closed surface is equal to the total charge enclosed by the surface.

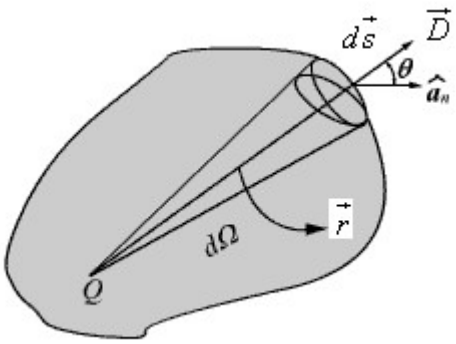


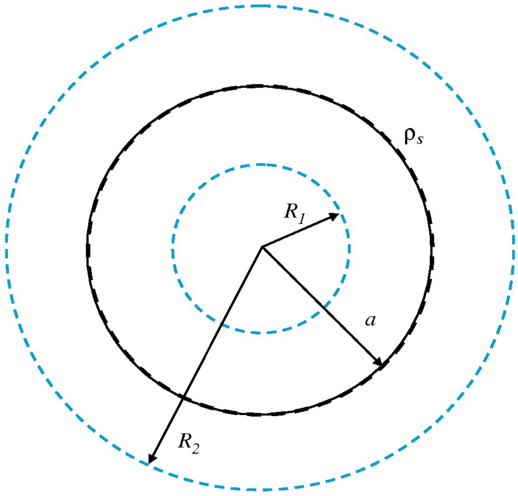
Figure 1.1: *Gauss's* Law

Application

Gauss's law is particularly useful in computing $\vec{E}$ or $\vec{D}$ where the charge distribution has some symmetry. illustrating the application of Gauss's Law with some examples include:

Example: A thin spherical shell of radius *a* carries uniform surface charge density ρ_s. Use Gauss's law to determine E.

Solution:



$$\oint_S \mathbf{D}\cdot d\mathbf{s} = Q$$

Symmetry suggests that **D** is radial in direction. Hence,

$$\mathbf{D} = \hat{\mathbf{R}} D_R$$
$$d\mathbf{s} = \hat{\mathbf{R}}\, ds$$
$$\oint_S \mathbf{D}\cdot d\mathbf{s} = \oint_S D_R\, ds = D_R(4\pi R^2) = Q$$
$$D_R = \frac{Q}{4\pi R^2}$$

- For a Gaussian surface of radius $R_1 < a$, no charge is enclosed. Hence, $Q = 0$, in which case $E = 0$.
- For a Gaussian surface of radius $R_2 > a$,

$$Q = \rho_s(4\pi a^2)$$

and

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon} = \frac{\hat{\mathbf{R}}}{\varepsilon} D_r = \frac{\hat{\mathbf{R}} Q}{4\pi\varepsilon R_2^2} = \hat{\mathbf{R}}\frac{4\pi\rho_s a^2}{4\pi\varepsilon R_2^2} = \hat{\mathbf{R}}\frac{\rho_s a^2}{\varepsilon R_2^2}\ .$$

<u>An infinite line charges</u>

Let's consider the problem of determination of the electric field produced by an infinite line charge of density r_LC/m. Let us consider a line charge positioned along the *z*-axis as shown in Fig. 1.2(a). Since the line charge is assumed to be infinitely long, the electric field will be of the form as shown in Fig. 1.2(b). If we consider a close cylindrical surface as shown in Fig. 1.2(a), using Gauss's theorem we can write,

$$\rho_L l = Q = \oint_S \varepsilon_0 \vec{E}.\vec{ds} = \int_{S_1} \varepsilon_0 \vec{E}.\vec{ds} + \int_{S_2} \varepsilon_0 \vec{E}.\vec{ds} + \int_{S_3} \varepsilon_0 \vec{E}.\vec{ds} \qquad (12)$$

Because the unit normal vector to areas S_1 and S_3 are perpendicular to the electric field, the surface integrals for the top and bottom surfaces evaluates to zero.

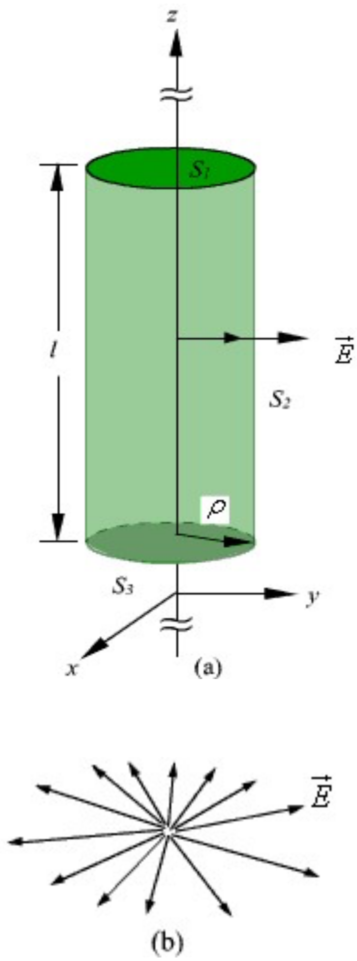


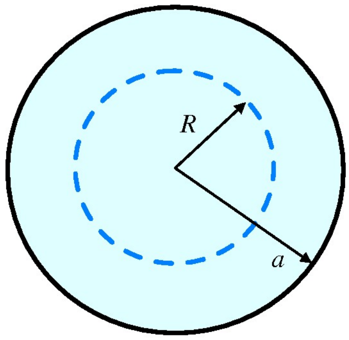
Fig 1.2: Infinite Line Charge

Hence, we can write,

$$\rho_L l = \varepsilon_0 E . 2\pi\rho l$$

$$\vec{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\hat{a}_\rho \quad (13)$$

Example: A spherical volume of radius a contains a uniform volume charge density ρv. Use Gauss's law to determine D for (a) R ≤ a and (b) R ≥ a.



$R < a$

For $R \le a$,

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \oint_S D_r \, ds = D_r(4\pi R^2)$$

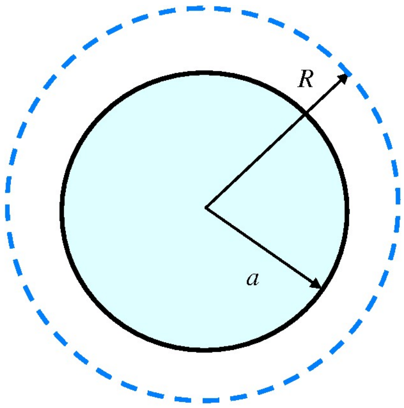
Q within a sphere of radius R is

$$Q = \frac{4}{3}\pi R^3 \rho_v$$

Hence,

$$4\pi R^2 D_R = \frac{4}{3}\pi R^3 \rho_v$$

$$D_r = \frac{\rho_v R}{3}, \qquad \mathbf{D} = \hat{\mathbf{R}} D_r = \hat{\mathbf{R}} \frac{\rho_v R}{3}, \qquad R \le a.$$



$R > a$

For $R \ge a$, total charge in sphere is

$$Q = \frac{4}{3}\pi a^3 \rho_v,$$

$$4\pi R^2 D_R = \frac{4}{3}\pi a^3 \rho_v,$$

$$\mathbf{D} = \hat{\mathbf{R}} D_r = \hat{\mathbf{R}} \frac{\rho_v a^3}{3R^2}, \qquad R \ge a.$$

Infinite Sheet of Charge

As a second example of application of Gauss's theorem, we consider an infinite charged sheet covering the *xz* plane as shown in fig. 1.3 Assuming a surface charge density of ρ_s for the infinite surface charge, if we consider a cylindrical volume having sides Δs placed symmetrically as shown in figure 2.5, we can write:

$$\oint_s \vec{D}\cdot d\vec{s} = 2D\Delta s = \rho_s \Delta s$$

$$\therefore \quad \vec{E} = \frac{\rho_s}{2\varepsilon_0}\hat{a}_y \qquad (14)$$

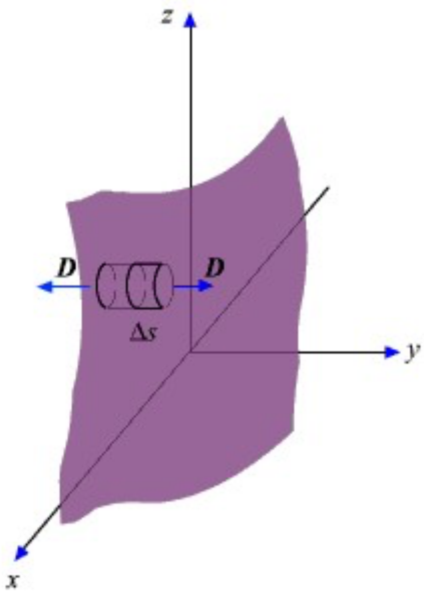


Fig 1.3: Infinite Sheet of Charge

It may be noted that the electric field strength is independent of distance. This is true for the infinite plane of charge; electric lines of force on either side of the charge will be perpendicular to the sheet and extend to infinity as parallel lines. As number of lines of force per unit area gives the strength of the field, the field becomes independent of distance. For a finite charge sheet, the field will be a function of distance.

Examples: READING ASSIGNMENT 1

As Gauss's law or Coulomb's law detailed: if the total flux through a closed surface equals the total charge enclosed by it, **Question**: What is $\vec{E}$**?**

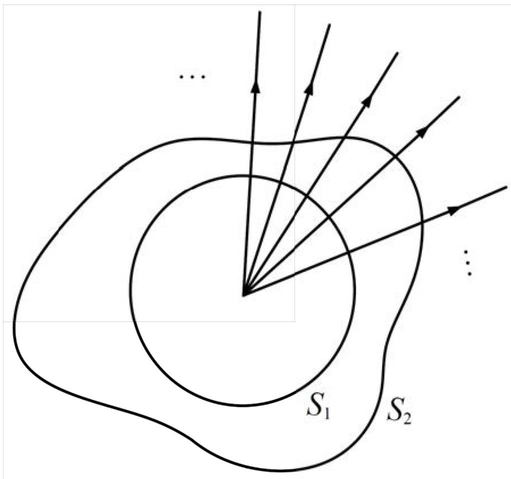


Fig. 1.4: Same amount of electric flux from a point charge passes through two surfaces S_1 and S_2.

As detailed and shown in the Gauss's LAW diagram of fig.1.3 above, this diagram shows the same amount of electric flux from a point charge passes through two surfaces S_1 and S_2.

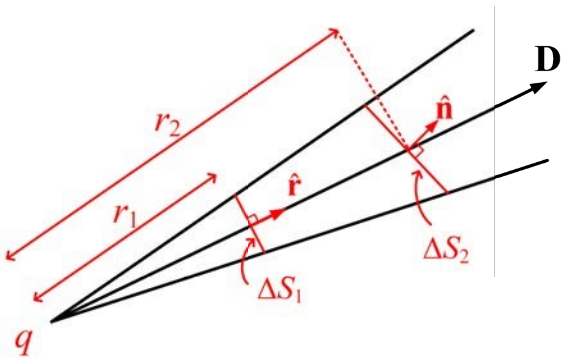


Figure 1.5: When a sliver of angular sector is taken, same amount of electric flux from a point charge passes through two incremental surfaces ΔS_1 and ΔS_2.

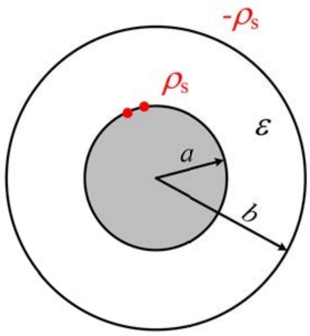


Figure 1.6: for a coaxial cylinder: Field between coaxial cylinders of unit length.

Solution Hint: Use symmetry and cylindrical coordinates as detailed in equation (12 - 13) and apply Gauss's law.

Examples: READING ASSIGNMENT 2

For the fields of a sphere of uniform charge density in Fig. 1.7 below. **Question**: What is $\vec{E}$**?**

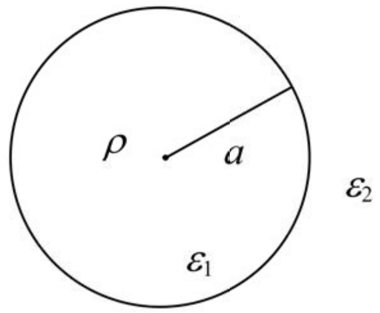


Fig. 1.7: Figure for reading assignment.

Solution Hint: Use symmetry and cylindrical coordinates as detailed in equation (12 - 13) and apply Gauss's law.

HND II: Electrical & Electronics Engineering
Course CODE: EEC 431
Course TITLE: Electromagnetic Field Theory

Course Lecturer:
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TOPIC 2: Principles & Applications of Static Magnetic Fields

PRINCIPLES & APPLICATIONS OF STATIC MAGNETIC FIELDS

Introduction to Magnetic Fields

Electrostatic fields are generated by static charges, magnetostatic fields are generated by static currents (charges that move with constant velocity in a particular direction).

•There are several similarities between electrostatic and magnetostatic fields

•For example, as we had E and D for electrostatics, we now use B and H to examine magnetic systems

•Our study of these fields allows us to evaluate and solve for a tremendous number of electric and electromechanical devices.

•Furthermore, this study, will provide the basis for formulating an universal theory of Electromagnetic Fields that is utilized in almost every aspect of electrical engineering.

Analysis between Electric & Magnetic Fields

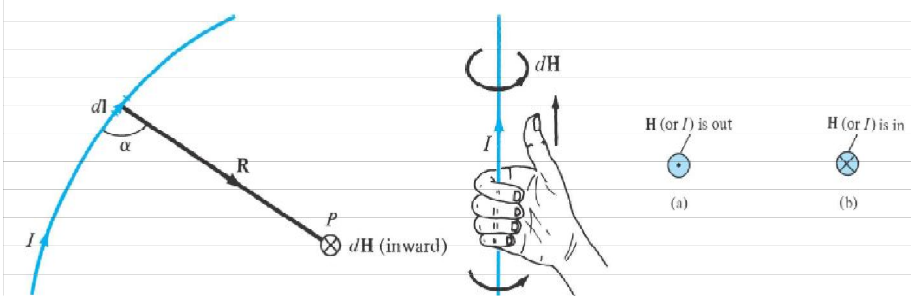
	Electric	Magnetic
• Basic Laws	$\vec{F} = \frac{Q_1 Q_1}{4\pi\varepsilon r^2}\hat{a}_r$	$d\vec{B} = \frac{\mu_0 I dl \times \hat{a}_r}{4\pi R^2}$
	$\oint \vec{D}\cdot d\vec{S} = Q_{enc}$	$\oint \vec{H}\cdot d\vec{l} = I_{enc}$
• Force Law	$\vec{F} = Q\bar{E}$	$\vec{F} = Q\vec{u}\times\vec{B}$
• Source Element	dQ	$Q\vec{u} = Id\vec{l}$
• Field intensity	$E = \frac{V}{l}(V/m)$	$H = \frac{I}{l}(A/m)$
• Flux density	$\bar{D} = \frac{\psi}{S}(C/m^2)$	$\vec{B} = \frac{\psi}{S}(Wb/m^2)$
• Relationship Between Fields	$\vec{D} = \varepsilon\bar{E}$	$\vec{B} = \mu\vec{H}$
• Potentials	$\bar{E} = -\nabla V$	$\vec{H} = -\nabla V_m, (\vec{J} = 0)$
	$V = \int \frac{\rho_L dl}{4\pi\varepsilon r}$	$A = \int \frac{\mu I d\vec{l}}{4\pi R}$
	$\psi = \oint \vec{D}\cdot d\vec{S}$	$\psi = \oint \vec{B}\cdot d\vec{S}$
• Flux	$\psi = Q = CV$	$\psi = LI$
	$I = C\frac{dV}{dt}$	$I = L\frac{dI}{dt}$
• Energy Density	$w_E = \frac{1}{2}\vec{D}\cdot\vec{E}$	$w_E = \frac{1}{2}\vec{B}\cdot\vec{H}$
• Poisson's Eqn.	$\nabla^2 V = -\frac{\rho_v}{\varepsilon}$	$\nabla^2 \vec{A} = -\mu J$

4

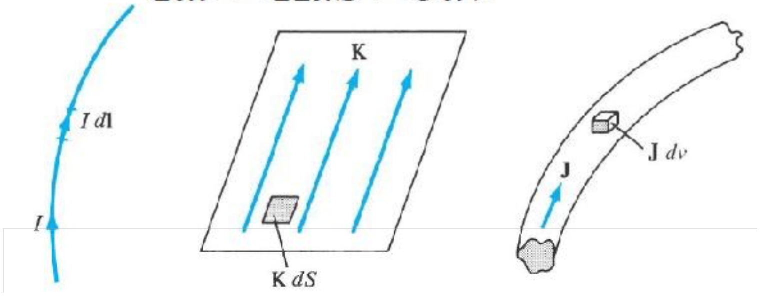
Biot-Savart's Law

The differential magnetic field intensity, $d\boldsymbol{H}$, produced at a point P, by the differential current element, $Id\boldsymbol{l}$, is proportional to the product $Id\boldsymbol{l}$ and the sine of the angle between the element and the line joining P to the element and is inversely proportional to the square of the distance, R, between P and the element

$$d\vec{H} = \frac{Id\vec{l} \times \hat{a}_R}{4\pi R^2} = \frac{Id\vec{l} \times \vec{R}}{4\pi R^3} = \frac{Idl \sin\alpha}{4\pi R^2}$$



$$Id\vec{l} = \vec{K}dS = \vec{J}dv$$



Considering different current distributions (as shown above) we can rewrite expression for field intensity as below,

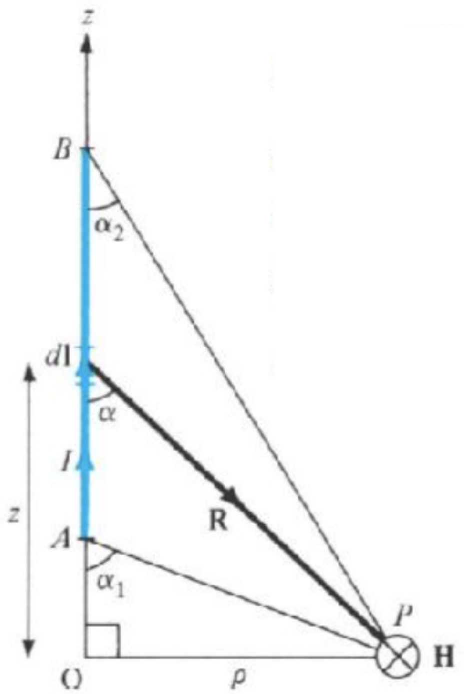
$$\vec{H} = \int_L \frac{Id\vec{l} \times \hat{a}_R}{4\pi R^2}$$

$$\vec{H} = \int_S \frac{\vec{K}dS \times \hat{a}_R}{4\pi R^2}$$

$$\vec{H} = \int_v \frac{\vec{J}dv \times \hat{a}_R}{4\pi R^2}$$

H **Field From a Strait Current Carrying Filament**

• The H field is determined for a strait filament of current in a manner very similar to that of the electric field determined from a line charge.



$$\vec{H} = \int_L \frac{I d\vec{l} \times \vec{R}}{4\pi R^3}$$

$$d\vec{l} = dz\hat{a}_z$$

$$\vec{R} = \rho\hat{a}_\rho - z\hat{a}_z$$

$$d\vec{l} \times \vec{R} = \rho dz\hat{a}_\phi$$

$$\vec{H} = \int_L \frac{I\rho dz\hat{a}_\phi}{4\pi(\rho^2 + z^2)^{3/2}}$$

Now,

$$z = \rho \cot \alpha$$

$$dz = -\rho(\csc^2 \alpha)d\alpha$$

$$\vec{H} = \int_L -\frac{I\rho^2(\csc^2 \alpha)d\alpha\hat{a}_\phi}{4\pi(\rho \csc \alpha)^3}$$

$$\vec{H} = -\frac{I}{4\pi\rho}\int_{\alpha_1}^{\alpha_2}(\sin \alpha)d\alpha\hat{a}_\phi$$

$$\vec{H} = \frac{I}{4\pi\rho}(\cos \alpha_2 - \cos \alpha_1)\hat{a}_\phi$$

$$\vec{H} = \frac{I}{4\pi\rho}\hat{a}_\phi \qquad \text{Line from } z = 0 \text{ to } \infty$$

$$\vec{H} = \frac{I}{2\pi\rho}\hat{a}_\phi \qquad \text{Line from } z = -\infty \text{ to } \infty$$

Using Biot-savart's law we can find the expression for field intensity due to different current carrying conductor configurations.

Ampere's law:

The line integral of $\boldsymbol{H}$ around a closed path is the same as the net current, I_{enc}, enclosed by the path,

$$\oint \vec{H} \cdot d\vec{l} = I_{enc}$$

– Similar to Gauss' law since Ampere's law is easily used to determine H when the current distribution is symmetrical.

– Ampere's law ALWAYS holds, even if the current distribution is NOT symmetrical, however the equation is typically used for symmetric cases.

– Like Gauss and Coulomb's Laws, Ampere's law is a special case of the Biot-Savart law and can be derived directly from it.

Derivation of Divergence's and Stoke's Theorem:

Recalled from the VECTOR introduction class the details on the following:

Divergence of a Vector:

- The divergence of a vector, **A**, at any given point P is the outward flux per unit volume as volume shrinks about P.

$$div\vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta v \to 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

Divergence Theorem:

- The divergence theorem states that the total outward flux of a vector field, **A**, through the closed surface, S, is the same as the volume integral of the divergence of **A**.
- This theorem is easily shown from the equation for the divergence of a vector field.

$$\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$$

$$div\vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta v \to 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

$$\int_v \nabla \cdot \vec{A} dv = \oint_s \vec{A} \cdot d\vec{S}$$

Stokes Theorem:

Stokes theorem states that the circulation of a vector field A, around a closed path, L is equal to the surface integral of the curl of A over the open surface S bounded by L. This theorem has been proven to hold as long as **A** and the curl of **A** are continuous along the closed surface S of a closed path L

- This theorem is easily shown from the equation for the curl of a vector field.

$$\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$$

$$curl\vec{A} = \nabla \times \vec{A} = \left(\lim_{\Delta S \to 0} \frac{\oint_L \vec{A} \cdot d\vec{l}}{\Delta S} \right)_{max} \hat{a}_n$$

$$\oint_L \vec{A} \cdot d\vec{l} - \oint_s \left(\nabla \times \vec{A}\right) \cdot d\vec{S}$$

Applying Stokes's theorem on Ampere's law

Applying Stokes's theorem on Ampere's law, we have,

$$I_{enc} = \oint_L \vec{H} \cdot d\vec{l} = \int_S \left(\nabla \times \vec{H}\right) \cdot d\vec{S}$$

We can also write using current density,

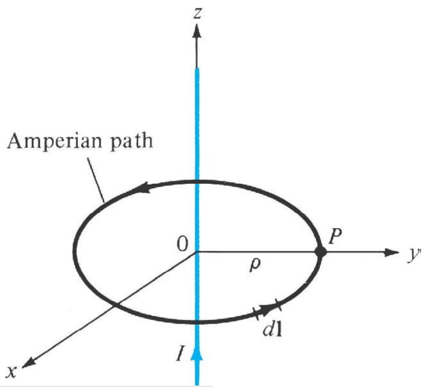
$$I_{enc} = \int_S \vec{J} \cdot d\vec{S}$$

So, from the above two equations,

$$\nabla \times \vec{H} = \vec{J}$$

Applications of Ampere's Circuit Law

- A simple application of Ampere's law can be used to easily derive the magnetic field intensity from an infinite line current ,



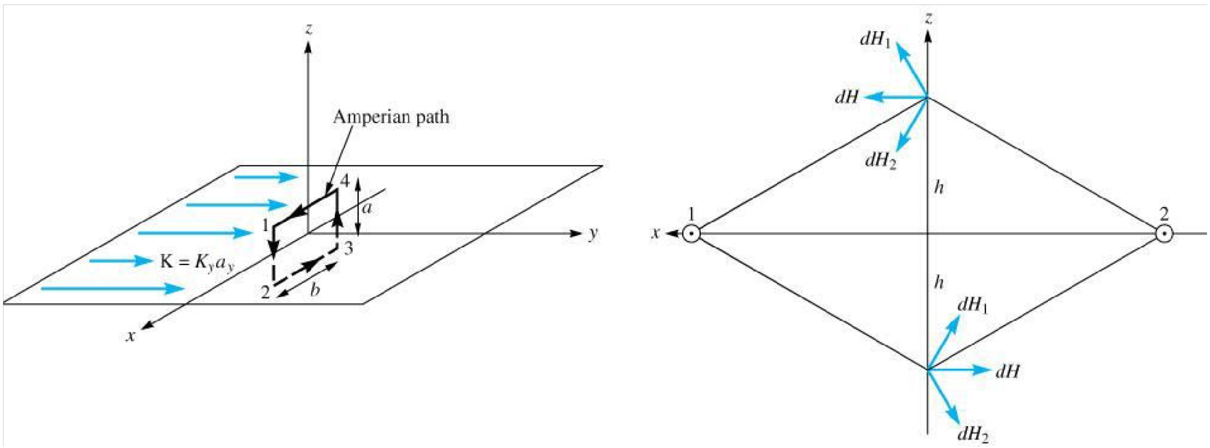
$$I_{enc} = \oint \vec{H} \cdot d\vec{l}$$

$$I = \int H_{\phi}\hat{a}_{\phi} \cdot \rho d\phi \hat{a}_{\phi} = H_{\phi}\int \rho d\phi = H_{\phi} \cdot 2\pi\rho$$

$$\vec{H} = \frac{I}{2\pi\rho}\hat{a}_{\phi}$$

Ampere's Circuit Law: Infinite Sheet of Current

- Consider an infinite sheet of current in the $z=0$ plane with a uniform current density, $\boldsymbol{K}=K_y\boldsymbol{a}_y$.



$$I_{enc} = \oint \vec{H} \cdot d\vec{l} = K_y b$$ Apply Ampere's law

$$\vec{H} = \begin{cases} H_0\hat{a}_x, z > 0 \\ -H_0\hat{a}_x, z < 0 \end{cases}$$

$$\oint \vec{H} \cdot d\vec{l} = \left(\int_1^2 + \int_2^3 + \int_3^4 + \int_4^1\right)\vec{H} \cdot d\vec{l}$$

$$= 0(-a) - H_0(-b) + 0(a) + H_0(b) = 2H_0 b$$

$from$ Ampere's law and integral summation

$H_o = K_y$

$$\vec{H} = \begin{cases} \frac{1}{2} K_y \hat{a}_x, z > 0 \\ -\frac{1}{2} K_y \hat{a}_x, z < 0 \end{cases}$$

Thus for an infinite sheet of charge,

$$\vec{H} = \frac{1}{2} \vec{K} \times \hat{a}_n$$

So, by using Ampere's circuital law, the expressions for magnetic field intensity of different structures can be derived.

Magnetic Flux Density

- Magnetic Flux density, **B**, is the magnetic equivalent of the electric flux density, **D**. As such, one can define,

$$\vec{B} = \mu_0 \vec{H}$$

where $\mu_0 = 4\pi \times 10^{-7} H / m$

- Similarly, Ampere's Law is,

$$I_{enc} - \oint \frac{\vec{B}}{\mu_0} \cdot d\hat{l}$$

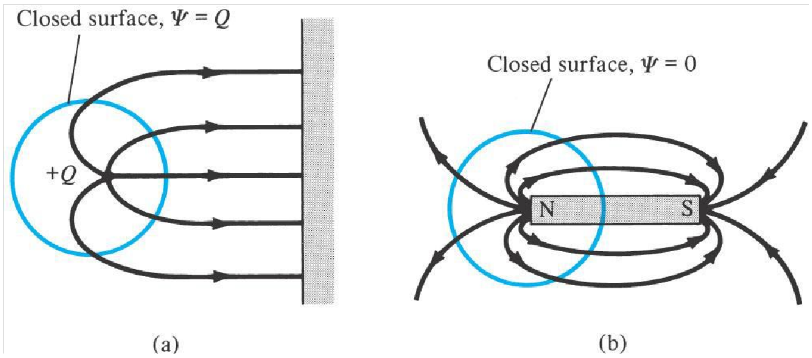
- And the Magnetic flux through a surface is,

$$\psi = \int_S \vec{B} \cdot d\vec{S} = \mu_0 \int_S \vec{H} \cdot d\vec{S}$$

- The magnetic flux through an enclosed system is,

$$\psi = \int_S \vec{B} \cdot d\vec{S} = \int_S \left(\nabla \cdot \vec{B}\right) dv$$

- Unlike electrostatic flux however, magnetic flux always follows a closed path and fold in on themselves. This simple statement has profound consequences. In electrostatics, we can easily define a point charge in which electric fields emanate to infinity. However, the solenoidal nature of the magnetic field requires magnetic flux to travel from a positive (north) to a negative (south) pole and it is not possible to have a single magnetic pole at any time.



–There are NO magnetic monopoles, stipulating that an isolated magnetic charge DOES NOT EXIST

–The minimum field requirement for magnetics is a dipole. So, mathematically,

$$\nabla \cdot \vec{B} = 0$$

Maxwell's Equation for Static Fields

Differential Form	Integral Form	Remarks
$\nabla \cdot \vec{D} = \rho_v$	$\oint_S \vec{D} \cdot d\vec{S} = \int \rho_v dv$	Gauss's Law
$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot d\vec{S} = 0$	Nonexistence of the Magnetic Monopole
$\nabla \times \vec{E} = 0$	$\oint_L \vec{E} \cdot d\vec{l} = 0$	Conservative nature of the Electric Field
$\nabla \times \vec{H} = \vec{J}$	$\oint_L \vec{H} \cdot d\vec{l} = \oint_S \vec{J} \cdot d\vec{S}$	Ampere's Law

Magnetic (Scalar & Vector) Potential

We can define a magnetic field using the following requirements.

$$\nabla \times \nabla V = 0$$
$$\nabla \cdot \left(\nabla \times \vec{A}\right) = 0$$

• Just as $\vec{E} = -\nabla V$, we can define a magnetic scalar potential Vm related to **H** when the current density is zero as

$$\vec{H} - -\nabla V_m, \vec{J} - 0$$
$$\vec{J} = \nabla \times \vec{H} = \nabla \times \left(-\nabla V_m\right) = 0$$
$$\nabla^2 V_m = 0, \vec{J} = 0$$

- The requirement for a solenoidal field (and Maxwell's 4th law of electrostatics) stipulates

$$\nabla \cdot \vec{B} = 0$$

- And we can therefore define a magnetic vector potential, **A**, as

$$\vec{B} = \nabla \times \vec{A}$$

- Just as we defined the Electric Potential as $V = \int \frac{dQ}{4\pi\varepsilon_0 r}$, We can define the Magnetic Vector Potential

as,

$$\vec{A} = \int_L \frac{\mu_0 I d\vec{l}}{4\pi R} \quad \text{for Line Current}$$

$$\vec{A} = \int_L \frac{\mu_0 \vec{K} dS}{4\pi R} \quad \text{for Surface Current}$$

$$\vec{A} = \int_L \frac{\mu_0 \vec{J} dv}{4\pi R} \quad \text{for Volume Current}$$

Forces Magnetic Field

Lorentz Force Law:

- The force on a charged particle in an electric field is simply **F**=q**E**
- However, in the presence of an electromagnetic field an additional force is imposed from the charge displacement of velocity, **u**, quantified by the magnetic field, **B**.
- The combined force is defined by Lorentz Force Law:

$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$$

- Equating the Lorentz force to Newton's force equation, we have,

$$\vec{F} - q(\vec{E} + \vec{u} \times \vec{B}) - m\vec{a} - m\frac{d\vec{u}}{dt}$$

Where 'a' is the acceleration of the particle is space.

Force on a Current Element

- One can also use field calculations to determine the force acting on a current element, $I d\boldsymbol{l} = \boldsymbol{K} dS = \boldsymbol{J} dv$, due to an applied external magnetic field $\boldsymbol{B}$.

Assume that a copper wire carries a current density, **J**= ρ_v**u**.

- We know:

$$I d\vec{l} = \vec{J} dv = \rho_v u = dQ u$$

- And that:

$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$$
$$where \quad \vec{E} \to 0$$
$$\vec{F} = q(\vec{u} \times \vec{B})$$

Thus, we can solve for the force on the first wire:

$$d\vec{F} = dq(\vec{u} \times \vec{B}) = I d\vec{l} \times \vec{B}$$
$$\vec{F} = \oint_L I d\vec{l} \times \vec{B}$$

$$Likewise$$
$$\vec{F} = \oint_L \vec{K} dS \times \vec{B} = \vec{F} = \oint_L \vec{J} dv \times \vec{B}$$

Examples: 1

An electron moving in the positive x-direction perpendicular to a magnetic field experiences a deflection in the negative z-direction. What is the direction of the magnetic field?

Solution: The magnetic force acting on a moving charged particle is

$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$$

,

$$q = -e$$
$$\mathbf{u} = \hat{\mathbf{x}} u$$
$$\mathbf{F}_m = -\hat{\mathbf{z}} F_m$$
$$-\hat{\mathbf{z}} F_m = -\hat{\mathbf{x}} ue \times \mathbf{B}$$

For the cross product to apply, B has to be in the positive *y*-direction.

Examples: 2

A proton moving with a speed of 2×10^6 m/s through a magnetic field with magnetic flux density of 2.5 T experiences a magnetic force of magnitude 4×10^{-13} N.
What is the angle between the magnetic field and the proton's velocity?

Solution:

$$
\begin{aligned}
F &= quB\sin\theta \\
\sin\theta &= \frac{F}{quB} \\
&= \frac{4\times10^{-13}}{1.6\times10^{-19}\times2\times10^{6}\times2.5} \\
&= 0.5
\end{aligned}
$$

$$\theta = \sin^{-1} 0.5 = 30^\circ \text{ or } 150^\circ.$$

Examples: 3

A charged particle with velocity u is moving in a medium containing uniform fields $E = \hat{x}E$ and $B = \hat{y}B$.
What should u be so that the particle experiences no net force on it?

Solution:

$$
\begin{aligned}
\mathbf{F}_e &= q\mathbf{E} = \hat{\mathbf{x}}qE \\
\mathbf{F}_m &= q\mathbf{u}\times\mathbf{B} = q(\mathbf{u}\times\hat{\mathbf{y}}B)
\end{aligned}
$$

For net force to be zero, F_m has to be along $-\hat{x}$, which requires u to be along $+\hat{z}$. Thus,

$$
\begin{aligned}
qE &= quB \\
u &= \frac{E}{B} \\
\mathbf{u} &= \hat{\mathbf{z}}\frac{E}{B}\,.
\end{aligned}
$$

If u also has a *y*-component, that component will exercise no force on the particle.

Examples: 4

A horizontal wire with a mass per unit length of 0.2 kg/m carries a current of 4 A in the +*x*-direction. If the wire is placed in a uniform magnetic flux density B, what should the direction and minimum magnitude of B be in order to magnetically lift the wire vertically upward? (Hint: The acceleration due to gravity is g = −zˆ9.8 m/s^2.)

Solution: For a length *l*,

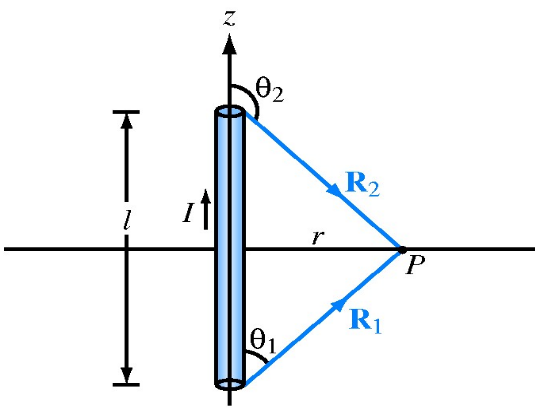
$$\mathbf{F}_g = -\hat{\mathbf{z}}0.2l \times 9.8 = -\hat{\mathbf{z}}1.96l \quad \text{(N)}$$
$$\mathbf{F}_m = \hat{\mathbf{x}}Il \times \mathbf{B}$$

For $F_m + F_g = 0$, F_m has to be along $+\hat{z}$, which means that B has to be along $+\hat{y}$. Hence,

$$1.96l = IlB$$
$$B = \frac{1.96}{I} = 0.49 \text{ (T), and}$$
$$\mathbf{B} = \hat{\mathbf{y}}0.49 \text{ (T).}$$

Examples: 5

A semi-infinite linear conductor extends between *z* = 0 and *z* = ∞ along the *z*-axis. If the current *I* in the conductor flows along the positive *z*-direction, find H at a point in the *x*–*y* plane at a radial distance *r* from the conductor.



$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi r}(\cos\theta_1 - \cos\theta_2)$$

For a conductor extending from $z = 0$ to $z = \infty$, $\theta_1 = 0$ and $\theta_2 = \pi$. Hence,

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi r}(1+1) = \hat{\boldsymbol{\phi}} \frac{I}{2\pi r} \quad \text{(A/m).}$$

HND II: Electrical & Electronics Engineering
Course CODE: EEC 431
Course TITLE: Electromagnetic Field Theory

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TOPIC 3: Principles & Applications of Time Varying Electromagnetic Fields

PRINCIPLES & APPLICATIONS OF TIME VARYING ELECTROMAGNETIC FIELDS

Inductors and Inductance:

- We now know that closed magnetic circuit carrying current I produces a magnetic field with flux

$$\Psi = \int \bar{B} \cdot d\bar{S}$$

- We define the flux linkage between a circuit with N identical turns as,

$$\lambda = N\Psi$$

- As long as the medium the flux passes through is linear (isotropic) then then flux linkage is proportional to the current I producing it and can be written as,

$$\lambda = LI$$

Where L is a constant of proportionality called the inductance of the circuit. A circuit that contains inductance is said to be an inductor.

- One can equate the inductance to the magnetic flux of the circuit as

$$L = \frac{\lambda}{I} = \frac{N\Psi}{I}$$

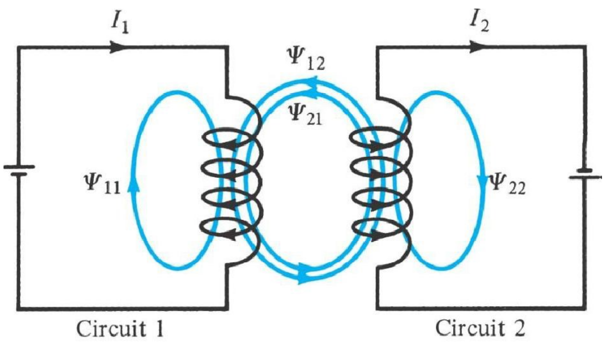
where L is measured in units of Henrys (H) = Wb/A.

- The magnetic energy (in Joules) stored by the inductor is expressed as,

$$W_m = \frac{1}{2} L I^2$$

Inductors and Inductance:

- Since we know that magnetic fields produce forces on nearby current elements, and that those magnetic fields can be generated by an isolated or coupled set of current carrying circuits, then it is only reasonable that such circuits may induce fields and magnetization between them.



• We can calculate the individual flux linkage between the two components as

$$\Psi_{12} = \int_{S_1} \vec{B}_2 \cdot d\vec{S}$$

• Likewise we can determine a mutual inductance between the circuits that is equal from circuit 12 as it is from circuit 21 as

$$M_{12} = \frac{\lambda_{12}}{I_2} = \frac{N_1 \Psi_{12}}{I_2}$$

• Individual inductances are

$$L_1 = \frac{\lambda_{11}}{I_1} = \frac{N_1 \Psi_1}{I_1} \qquad L_2 = \frac{\lambda_{22}}{I_2} = \frac{N_2 \Psi_2}{I_2}$$

• The total magnetic energy in the circuit is

$$W_m = W_1 + W_2 + W_{12} = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 \pm M_{12} I_1 I_2$$

• Mutual inductance may be calculated by the following method,

– Determine the internal inductance, Lin for the flux generated by the first inductor

– Determine the external inductance, Lext produced by the flux external of the first inductor

– The sum of the internal and external inductance equals the individual inductances plus the mutual inductance between the elements

$$M_{12} = \frac{\lambda_{12}}{I_2} = \frac{N_1 \Psi_{12}}{I_2} \qquad \Psi_{12} = \int_{S_1} \vec{B}_2 \cdot d\vec{S}$$

Magnetic Energy

We can derive a similar term as derived for electric energy, for magnetic energy using the relation for energy as a function of inductance.

$$W_m = \frac{1}{2} L I^2$$

$$\Delta L = \frac{\Delta\Psi}{\Delta I} = \frac{\mu H \Delta x \Delta y}{\Delta I}$$

$$where, \Delta I = H \Delta z$$

$$\Delta L = \frac{\mu H \Delta x \Delta y}{\Delta I} = \frac{\mu \Delta x \Delta y}{\Delta z}$$

$$\Delta W_m = \frac{1}{2} \Delta L \Delta I^2 = \frac{1}{2} \mu H^2 \Delta x \Delta y \Delta z$$

$$\Delta W_m = \frac{1}{2} \mu H^2 \Delta v$$

$$w_m = \lim_{\Delta v \to 0} \frac{\Delta W_m}{\Delta v} = \frac{1}{2} \mu H^2 = \frac{1}{2}\left(\vec{H} \cdot \vec{B}\right) = \frac{B^2}{2\mu}$$

$$W_m = \int w_m dv = \int \frac{1}{2}\left(\vec{H} \cdot \vec{B}\right) dv = \int \frac{1}{2} \mu H^2 dv$$

Magnetic Circuits

•The following relations allow one to solve magnetic field problems in a manner similar to that of electronic circuits. It provides a clear means of designing transformers, motors, generators, and relays using a lumped circuit model. The analogy between electronic and magnetic circuits is provided below.

Table 8.4 Analogy between Electric and Magnetic Circuits

Electric	Magnetic
Conductivity σ	Permeability μ
Field intensity E	Field intensity H
Current $I = \int \mathbf{J} \cdot d\mathbf{S}$	Magnetic flux $\Psi = \int \mathbf{B} \cdot d\mathbf{S}$
Current density $J = \frac{I}{S} = \sigma E$	Flux density $B = \frac{\Psi}{S} = \mu H$
Electromotive force (emf) V	Magnetomotive force (mmf) $\mathcal{F}$
Resistance R	Reluctance $\mathcal{R}$
Conductance $G = \frac{1}{R}$	Permeance $\mathcal{P} = \frac{1}{\mathcal{R}}$
Ohm's law $R = \frac{V}{I} = \frac{\ell}{\sigma S}$ or $V = E\ell = IR$	Ohm's law $\mathcal{R} = \frac{\mathcal{F}}{\Psi} = \frac{\ell}{\mu S}$ or $\mathcal{F} = H\ell = \Psi\mathcal{R} = NI$
Kirchhoff's laws: $\Sigma I = 0$, $\Sigma V - \Sigma RI = 0$	Kirchhoff's laws: $\Sigma \Psi = 0$, $\Sigma \mathcal{F} - \Sigma \mathcal{R} \Psi = 0$

Faraday's Law for induced emf:

Faraday's law state that… the Induced electromotive force (emf) (in volts) in any closed circuit is equal to the time rate of change of magnetic flux by the circuit,

$$V_{emf} = -\frac{d\lambda}{dt} = -N\frac{d\Psi}{dt}$$

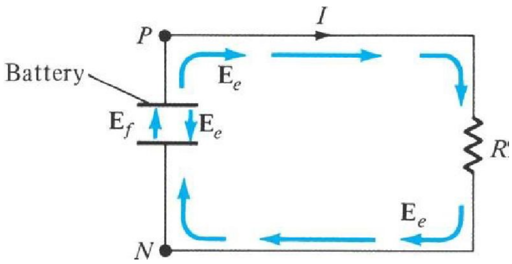
Where: λ is the flux linkage,
ψ is the magnetic flux,
N is the number of turns in the inductor, and
t represents a time interval.

The negative sign shows that the induced voltage acts to oppose the flux producing it.

•The statement above is known as Lenz's Law: the induced voltage acts to oppose the flux producing it.

•Examples of emf generated electric fields: electric generators, batteries, thermocouples, fuel cells, photovoltaic cells, transformers.

• To elaborate on emf, let's consider a battery circuit.



• Electrochemical action within the battery results and emf produced electric field, **Ef**

• Acuminated charges at the terminals provide an electrostatic field **Ee** that also exist that counteracts the emf generated potential.

$$\vec{E} = \vec{E}_f + \vec{E}_e$$

$$\oint_L \vec{E}\cdot d\vec{l} = \oint_L \vec{E}_f \cdot d\vec{l} + 0 = \int_N^P \vec{E}_f \cdot d\vec{l}$$

• The total emf generated in the between the two open terminals in the battery is therefore,

$$V_{emf} = \int_N^P \vec{E}_f \cdot d\vec{l} = -\int_N^P \vec{E}_e \cdot d\vec{l} = IR$$

Transformer and Motional Electromotive Forces:

- The variation of flux with time may be caused by three ways

  1. Having a stationary loop in a time-varying **B** field
  2. Having a time-varying loop in a static **B** field
  3. Having a time-varying loop in a time-varying **B** field

- A stationary loop in a time-varying **B** field

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$$

- A time-varying loop in a static **B** field

$$\nabla \times \vec{E}_m = \nabla \times \left(\vec{u} \times \vec{B}\right)$$

- A time-varying loop in a time-varying **B** field

$$\nabla \times \vec{E}_m = -\frac{d\vec{B}}{dt} + \nabla \times \left(\vec{u} \times \vec{B}\right)$$

Displacement Current:

- Lets now examine time dependent fields from the perspective on Ampere's Law.

$$\nabla \times \vec{H} = \vec{J}$$

$$\nabla \cdot \left(\nabla \times \vec{H}\right) = 0 = \nabla \cdot \vec{J}$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t} \neq 0$$

This vector identity for the cross product is mathematically valid. However, it requires that the continuity eqn. equals zero, which is not valid from an electrostatics standpoint!

$$\nabla \times \vec{H} = \vec{J} + \vec{J}_d$$

Thus, lets add an additional current density term to balance the electrostatic field requirement

$$\nabla \cdot \left(\nabla \times \vec{H}\right) = 0 = \nabla \cdot \vec{J} + \nabla \cdot \vec{J}_d$$

$$\nabla \cdot \vec{J}_d = -\nabla \cdot \vec{J} = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t}\left(\nabla \cdot \vec{D}\right) = \nabla \cdot \frac{\partial \vec{D}}{\partial t}$$

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

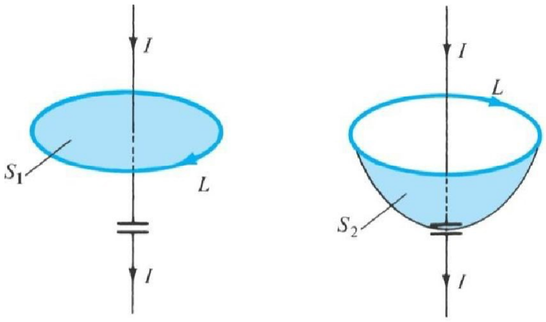
We can now define the displacement current density as the time derivative of the displacement vector

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Another of Maxwell's for time varying fields

This one relates Magnetic Field Intensity to conduction and displacement current densities

- We can apply the displacement current concept on the simple case of a capacitive element in a simple electronic circuit, as shown below.



Based on the equation for displacement current density, we can define the displacement current in a circuit as shown. Applying Ampere's circuit law to a closed path provides the following eqn. for current on the first side of the capacitive element. However, surface 2 is the opposite side of the capacitor and has no conduction current allowing for no enclosed current at surface 2. If $J = 0$ on the second surface, then Jd must be generated on the second surface to create a time displaced current equal to current on surface 1.

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$I_d = \int \vec{J} \cdot d\vec{S} = \int \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$$

$$\oint_L \vec{H} \cdot d\vec{l} = \int_{S_1} \vec{J} \cdot d\vec{S} = I_{enc} = I$$

$$\oint_L \vec{H} \cdot d\vec{l} = \int_{S_2} \vec{J} \cdot d\vec{S} = I_{enc} = 0$$

If J =0 on the second surface, then J_d must be generated on the second surface to create a time displaced current equal to current on surface 1.

$$\oint_L \vec{H} \cdot d\vec{l} = \int_{S_2} \vec{J}_d \cdot d\vec{S} = \frac{\partial}{\partial t} \int_{S_2} \vec{D} \cdot d\vec{S} = \frac{dQ}{dt} = I$$

We know,

$$D = \varepsilon E = \varepsilon \frac{V}{d}$$

So,

$$J_d = \frac{\partial D}{\partial t} = \frac{\varepsilon}{d}\frac{dV}{dt}$$

$$\Rightarrow I_d = \vec{J}_d \cdot \vec{S} = \frac{\varepsilon S}{d}\frac{dV}{dt} = C\frac{dV}{dt}$$

$$from\_surface\_1$$

$$I_c = \frac{dQ}{dt} = S\frac{d\rho_s}{dt} = S\frac{dD}{dt} = \varepsilon S\frac{dE}{dt} = \frac{\varepsilon S}{d}\frac{dV}{dt} = C\frac{dV}{dt}$$

Maxwell's Time Dependent Equations

The Maxwell's equations for time dependent fields are,

Differential Form	Integral Form	Remarks
$\nabla \cdot \vec{D} = \rho_v$	$\oint_S \vec{D} \cdot d\vec{S} = \int \rho_v dv$	Gauss's Law
$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot d\vec{S} = 0$	Nonexistence of the Magnetic Monopole
$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint_L \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t}\int_S \vec{B} \cdot d\vec{S}$	Faraday's Law
$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\oint_L \vec{H} \cdot d\vec{l} = \oint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{S}$	Ampere's Circuit Law

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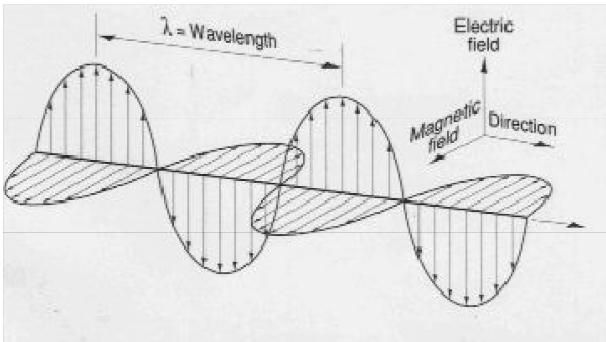
TOPIC 4: Principles & Applications of Plane Waves

PRINCIPLES & APPLICATIONS OF PLANE WAVES

Plane Wave:

A plane wave only has dependence (and, therefore, propagation) in one direction. In the figure below, the plane wave is propagating in the z direction. Notice that the polarization is in the y direction (thus, it is $Ey(z)$), but the wave is propagating only in the z direction.



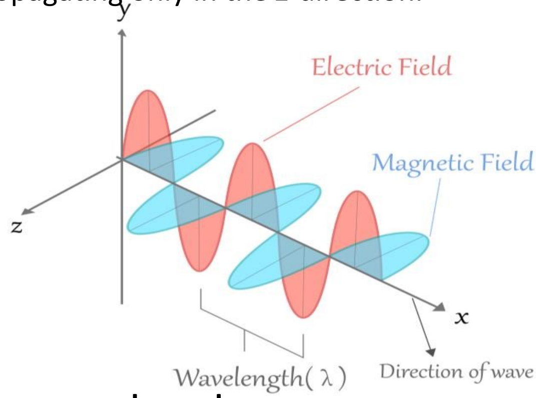


Figure 1. A plane wave $E_y(z)$ has no x or y dependence.

Wave Equations:

We define the **wavelength** λ to be the distance between successive peaks. Since the wave has no dependence in the x or y direction, we can simplify the description of the wave as follows.

$$E_y(x, y, z, t) = E_y(z, t) \tag{Equ. 1}$$

If the wave time harmonic is known, the z and t dependence can be separated as follows:

$$E_y(z, t) = E_y(z)e^{j\omega t} \tag{Equ. 2}$$

Let's investigate whether (and how) this function will satisfy the wave equation. First, we'll determine the second derivative of E_y with respect to z:

$$\frac{\partial^2 E_y(z, t)}{\partial z^2} = \frac{\partial^2 E_y(z)}{\partial z^2}e^{j\omega t} \tag{Equ. 3}$$

Next, we'll calculate the second derivative with respect to t:

$$\frac{\partial^2 E_y(z, t)}{\partial t^2} = E_y(z)(j\omega)^2 e^{j\omega t} \tag{Equ. 4}$$

Substituting Equations 3 and 4 into the wave equation 2 above, we find:

$$\frac{\partial^2 E_y(z)}{\partial z^2} e^{j\omega t} - \frac{1}{c^2} E_y(z)(j\omega)^2 e^{j\omega t} = 0$$ (Equ. 5)

This can be simplified as follows:

$$\frac{\partial^2 E_y(z)}{\partial z^2} + \left(\frac{\omega}{c}\right)^2 E_y(z) = 0$$ (Equ. 6)

If we define k (called the **wave number**) as follows:

$$k = \frac{\omega}{c}$$ (Equ. 7)

Then Equation 6 simplifies to:

$$\frac{\partial^2 E_y(z)}{\partial z^2} + k^2 E_y(z) = 0$$ (Equ. 8)

This is known as the **one-dimensional Helmholtz equation**, and it is a well-known differential equation with a well-known solution:

$$E_y(z) = ae^{-jkz} + be^{jkz}$$ (Equ. 9)

Adding the time dependence back into this result, we obtain:

$$E_y(z,t) = ae^{-jkz}e^{j\omega t} + be^{jkz}e^{j\omega t}$$ (Equ. 10)

Which can be written as:

$$E_y(z,t) = ae^{j(\omega t - kz)} + be^{j(\omega t + kz)}$$ (Equ. 11)

Notice that this function has the form of Equation 2 as long as Equation 1 is followed, so it is guaranteed to be a solution to the wave equation. In order to ensure that Equation 11 is a solution to the wave equation, the following relationship must be true.

$$v_\phi = \frac{dz}{dt} = \pm c = \pm\frac{\omega}{k}$$ (Equ. 12)

The positive signs correspond to a right-moving wave, and the negative signs to a left-moving wave. We now have several variables that describe the time and spatial variation of

$E(z,t)$: wavelength λ, wave number k, radial frequency ω, frequency f, period T, and phase velocity $v\phi$. Figure 2 below is designed to graphically demonstrate the relationships among these six variables. If you know two of the variables, you will be able to calculate the other four.